

Weibull Distribution

General Definitions

The Weibull distribution aims to fix some of the shortcomings of the exponential distribution. It removes the *memoryless* characteristic, by adding a *shape factor* β that models the evolution of failure rates over time.

It has this PDF:

$$f(t, \eta, \beta) = \begin{cases} \frac{\beta}{\eta} \left(\frac{t}{\eta}\right)^{\beta-1} e^{-(t/\eta)^\beta} & \text{if } t \geq 0 \\ 0 & \text{if } t < 0 \end{cases}$$

And the CDF:

$$F(t) = P[T \leq t] = 1 - e^{-(t/\eta)^\beta}$$

The η parameter is called the *characteristic life* or *scale parameter*, which represents the time where 63.2% of the population has failed, regardless of the value of β . It comes from evaluating the CDF at $t = \eta$.

$$F(\eta) = 1 - e^{-1} \approx 63.2\%$$